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## Earned Value Management in a Failing Software Development Project

*How EVM Metrics Drive Corrective Action and Termination Decisions*

### CASE STUDY

**Project: CloudSync ERP Platform (2022–2024)**

Industry: Logistics & Supply Chain Technology

Module 2 – BL: 5 (Evaluate)

|                 |                                |
|-----------------|--------------------------------|
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| Course Code     | SWE2028                        |

1. Introduction

Software development projects are inherently complex undertakings, fraught with technical uncertainty, evolving requirements, and resource constraints. Yet one of the most persistent challenges for project managers is not the complexity itself, but the delayed recognition of project distress. By the time a project is visibly failing, corrective action is often far more expensive — or outright impossible — than it would have been at the first sign of trouble.

Earned Value Management (EVM) is a structured, integrated performance measurement methodology that provides objective, quantitative data on both cost and schedule performance simultaneously. Unlike simple budget tracking (which only reveals how much has been spent) or milestone tracking (which only reveals whether deliverables are complete), EVM ties expenditure directly to the value of completed work. This makes it uniquely powerful for detecting project distress early and for supporting rational decisions about corrective action or project termination.

This case study examines the CloudSync ERP Platform project, a large-scale software initiative undertaken by a mid-sized logistics company between 2022 and 2024. The project experienced significant schedule delays and cost overruns that went undetected for several months. Through retrospective EVM analysis, this study evaluates how EVM metrics could have detected the crisis earlier, guided corrective actions, and ultimately supported the decision-making framework that led to the project’s partial recovery.

2. Project Background and Context

2.1 Organisation and Strategic Motivation

FreightFlow Logistics, a mid-sized third-party logistics (3PL) company operating across Southeast Asia, initiated the CloudSync ERP Platform project in January 2022. The organisation managed a network of 1,200+ clients, 8 regional warehouses, and daily shipments exceeding 45,000 packages. Their legacy ERP system, deployed in 2009, was no longer capable of handling real-time inventory visibility, AI-driven route optimisation, or integration with modern e-commerce platforms.

The strategic drivers for the project included:

- Replacing the outdated ERP with a cloud-native microservices platform
- Enabling real-time inventory visibility across all eight warehouse nodes
- Integrating with 200+ e-commerce and courier APIs
- Supporting predictive demand forecasting using machine learning
- Complying with updated data localisation regulations across three countries

2.2 Project Scope and Baseline Plan

The project was formally approved with the following baseline parameters:

| Parameter          | Detail                                              |
|--------------------|-----------------------------------------------------|
| Project Duration   | 24 months (Jan 2022 – Dec 2023)                     |
| Total Budget (BAC) | \$1,200,000 USD                                     |
| Team Size          | 18 FTEs (Dev, QA, DevOps, UX, PM, Architect)        |
| Delivery Model     | Agile-Waterfall Hybrid (Phases with Sprints)        |
| Reporting Cadence  | Monthly status reports (no formal EVM at inception) |

### 2.3 Phase Breakdown and Planned Value (PV)

The budget was distributed across four project phases:

| Phase                                | Duration     | Planned Budget | Cumulative PV |
|--------------------------------------|--------------|----------------|---------------|
| Phase 1: Requirements & Architecture | Months 1–5   | \$240,000      | \$240,000     |
| Phase 2: Core Development            | Months 6–14  | \$600,000      | \$840,000     |
| Phase 3: Integration & Testing       | Months 15–20 | \$264,000      | \$1,104,000   |
| Phase 4: UAT & Go-Live               | Months 21–24 | \$96,000       | \$1,200,000   |

### 2.4 Risk Register Summary at Project Inception

The project's initial risk register identified the following key risks, all of which were rated 'medium' and placed on a watch list without active mitigation plans:

- Technical risk: No prior microservices or cloud-native development within the team
- Regulatory risk: Data localisation laws under review in Malaysia and Indonesia
- Resource risk: Shortage of experienced cloud architects in the regional talent market
- Integration risk: Dependency on third-party API documentation from 200+ vendors
- Schedule risk: Executive-mandated December 2023 go-live date

## 3. EVM Fundamentals: The Analytical Framework

### 3.1 Core EVM Metrics

Earned Value Management is built on three core data points, from which all performance indicators are derived:

| Metric        | Abbreviation | Definition                                                         |
|---------------|--------------|--------------------------------------------------------------------|
| Planned Value | PV           | Budgeted cost of work scheduled to be done by the measurement date |
| Earned Value  | EV           | Budgeted cost of work actually completed by the measurement date   |
| Actual Cost   | AC           | Actual money spent to accomplish the completed work                |

### 3.2 Derived Performance Indices and Forecasts

| Indicator                        | Formula                   | Interpretation                        | Alert Threshold       |
|----------------------------------|---------------------------|---------------------------------------|-----------------------|
| Cost Performance Index (CPI)     | $EV / AC$                 | $> 1.0$ = under budget                | $< 0.85$ = Warning    |
| Schedule Performance Index (SPI) | $EV / PV$                 | $> 1.0$ = ahead of schedule           | $< 0.85$ = Warning    |
| Estimate at Completion (EAC)     | $BAC / CPI$               | Projected total project cost          | N/A                   |
| Variance at Completion (VAC)     | $BAC - EAC$               | Negative = over-budget at end         | $< 0$ = concern       |
| Estimate to Complete (ETC)       | $EAC - AC$                | Cost needed to finish remaining work  | N/A                   |
| To-Complete Perf. Index (TCPI)   | $(BAC - EV) / (BAC - AC)$ | Efficiency needed to finish on budget | $> 1.1$ = unrealistic |

These metrics, when tracked consistently over time, create a trend picture that is far more diagnostic than any single data point. A CPI that begins at 0.90 and progressively falls to 0.65 over four months tells a very different story than one that drops sharply in a single month due to a one-time event.

## 4. Project Performance Data: Month-by-Month Analysis

### 4.1 Performance Metrics Table (Months 1–16)

The following table presents the reconstructed EVM data for the CloudSync project. Note that formal EVM tracking was only introduced at Month 10; data for Months 1–9 was reconstructed retrospectively through timesheet analysis and sprint velocity reviews.

| Month | Phase | PV (\$k) | EV (\$k) | AC (\$k) | CPI  | SPI  | % Done | Status               |
|-------|-------|----------|----------|----------|------|------|--------|----------------------|
| 1     | Ph.1  | 50       | 46       | 55       | 0.84 | 0.92 | 4%     | Early Warning        |
| 2     | Ph.1  | 100      | 78       | 108      | 0.72 | 0.78 | 9%     | Red Flag             |
| 3     | Ph.1  | 150      | 115      | 162      | 0.71 | 0.77 | 12%    | Red Flag             |
| 4     | Ph.1  | 200      | 148      | 215      | 0.69 | 0.74 | 16%    | Red Flag             |
| 5     | Ph.1  | 240      | 192      | 272      | 0.71 | 0.80 | 19%    | Red Flag             |
| 6     | Ph.2  | 340      | 258      | 395      | 0.65 | 0.76 | 25%    | Critical             |
| 7     | Ph.2  | 440      | 318      | 540      | 0.59 | 0.72 | 31%    | Critical             |
| 8     | Ph.2  | 540      | 380      | 720      | 0.53 | 0.70 | 35%    | Critical             |
| 9     | Ph.2  | 640      | 445      | 920      | 0.48 | 0.70 | 40%    | Crisis Point         |
| 10    | Ph.2  | 740      | 520      | 1,100    | 0.47 | 0.70 | 45%    | EVM Introduced       |
| 11    | Ph.2  | 840      | 620      | 1,240    | 0.50 | 0.74 | 50%    | Recovery Plan Active |
| 12    | Ph.2  | 840      | 730      | 1,350    | 0.54 | 0.87 | 56%    | Improving            |

|    |        |       |       |       |      |      |     |                    |
|----|--------|-------|-------|-------|------|------|-----|--------------------|
| 13 | Ph.2–3 | 840   | 820   | 1,460 | 0.56 | 0.98 | 64% | Improving          |
| 14 | Ph.3   | 940   | 920   | 1,580 | 0.58 | 0.98 | 72% | Stabilised         |
| 15 | Ph.3   | 1,040 | 1,000 | 1,680 | 0.60 | 0.96 | 81% | On Track (Revised) |
| 16 | Ph.3–4 | 1,104 | 1,085 | 1,760 | 0.62 | 0.98 | 88% | On Track (Revised) |

## 4.2 Key Observations

Several critical patterns emerge from the performance data:

- **CPI Collapse (Months 1–9):** The CPI declined persistently from 0.84 at Month 1 to 0.47 at Month 10, never once recovering. A persistent downward CPI trend across three or more months is one of the strongest predictors of project failure.
- **SPI Stagnation:** The SPI hovered between 0.70 and 0.80 for the first 11 months, indicating a consistent delivery rate of only 70–80% of planned work each period.
- **Delayed Recognition:** Without formal EVM, the project team was reporting ‘on track’ through Month 6 based on milestone status reports, which showed phases ‘in progress’ without revealing the true cost-to-completion picture.
- **Recovery Signal:** From Month 12 onward, both CPI and SPI show consistent (if modest) improvement following corrective action, validating the recovery intervention.

## 5. EVM Forecasting: The Case for Early Intervention

### 5.1 What Month 3 Data Would Have Revealed

Had EVM been formally implemented from project inception, the Month 3 dashboard would have shown:

| Metric          | Value at Month 3 | Implication                                                                         |
|-----------------|------------------|-------------------------------------------------------------------------------------|
| CPI             | 0.71             | Only \$0.71 of work done per \$1 spent                                              |
| EAC (BAC / CPI) | \$1,690,000      | 41% projected cost overrun already visible                                          |
| VAC             | –\$490,000       | Forecasted \$490k overrun at completion                                             |
| TCPI            | 1.10             | Would require 55% improvement in efficiency to meet budget — borderline unrealistic |

A TCPI of 1.10 at Month 3 is a critical threshold. Research in project management literature consistently finds that CPI values established by Month 3 rarely improve by more than 0.10 in practice. The Month 3 EAC forecast of \$1.69M, had it been visible to leadership, would have been grounds for an immediate project review and re-planning exercise.

## 5.2 Month 9 Crisis Point: Full EVM Analysis

The Month 9 data represents the first point at which formal EVM metrics were calculated by a newly assigned PMO consultant. The results were alarming:

| EVM Calculation                  | Formula & Value                         | Decision Significance                              |
|----------------------------------|-----------------------------------------|----------------------------------------------------|
| Budget at Completion (BAC)       | \$1,200,000                             | Original approved budget                           |
| Actual Cost to Date (AC)         | \$920,000                               | 77% of budget spent at 37.5% of timeline           |
| Earned Value (EV)                | \$445,000                               | Work completed worth only 37% of budget            |
| $CPI = EV / AC$                  | 0.48                                    | For every \$1 spent, only \$0.48 of work delivered |
| $EAC = BAC / CPI$                | \$2,500,000                             | Projected final cost = 108% overrun                |
| $VAC = BAC - EAC$                | -\$1,300,000                            | Forecast to exceed budget by \$1.3M                |
| $ETC = EAC - AC$                 | \$1,580,000                             | \$1.58M still needed to complete the project       |
| $TCPI = (BAC - EV) / (BAC - AC)$ | $(755,000) / (-280,000) =$<br>UNDEFINED | Budget fully consumed; original BAC unachievable   |

The undefined TCPI — resulting from AC already exceeding BAC — is perhaps the most striking indicator. It mathematically confirms that the project cannot be completed within its original approved budget under any efficiency scenario. This is not merely a warning sign; it is a definitive trigger for formal project review under any governance framework.

## 6. Root Cause Analysis of Performance Degradation

The EVM data reveals what happened, but root cause analysis explains why. Post-mortem interviews and sprint retrospective data identified five primary causes:

### 6.1 Technology Inexperience (Contributing Factor: 38%)

The development team had no prior experience with cloud-native microservices architecture. The organisation had historically delivered monolithic enterprise applications. The architecture selected for CloudSync (Kubernetes-orchestrated microservices on AWS) required the team to learn container orchestration, service mesh patterns, and event-driven design concurrently with delivery. This resulted in three major architectural redesigns in Months 2, 5, and 8, each consuming 3–4 weeks of rework.

An architectural proof-of-concept (spike) during pre-project planning would have exposed these gaps and enabled either team training or scope simplification before commitments were made.

## 6.2 Uncontrolled Scope Additions (Contributing Factor: 27%)

The regulatory landscape in Malaysia and Indonesia changed during the project. New data localisation requirements mandated that all customer financial data be stored within national borders, requiring a complete redesign of the data tier that had already been partially built. Additionally, three business units requested feature additions totalling an estimated 320 hours of development effort, none of which were formally scoped or budgeted.

Without a Change Control Board and formal change request process, these additions were absorbed into the project without adjusting the budget baseline or the performance measurement baseline, making EVM comparisons against the original plan increasingly misleading.

## 6.3 Resource Attrition and Replacement Lag (Contributing Factor: 19%)

The lead cloud architect resigned at Month 5 to join a hyperscaler, and a senior backend developer was poached at Month 7. Combined ramp-up time for replacements exceeded 6 weeks. Knowledge transfer was inadequate due to the departure of institutional knowledge without documentation. This created duplicate work in the authentication module and delayed the API integration framework by approximately three weeks.

## 6.4 Integration Complexity with Third-Party APIs (Contributing Factor: 10%)

Of the 200 planned API integrations, 43 had outdated, incorrect, or non-existent documentation. Testing these integrations revealed data format mismatches and authentication protocol differences that had not been accounted for in the original estimate. Each failed integration required an average of 18 additional hours of engineering work.

## 6.5 Inadequate Testing Infrastructure (Contributing Factor: 6%)

The QA environment was provisioned on shared infrastructure, leading to test execution delays and environment instability. Regression test cycles planned for 5 days routinely required 9–12 days. This bottleneck, though individually modest, compounded across every development sprint and contributed to the persistent SPI underperformance.

## 7. Decision Framework: Continue, Re-Scope, or Terminate

When the Month 10 EVM analysis was presented to the executive steering committee, the fundamental question was binary: should the organisation continue investing in CloudSync, or should the project be terminated?

The committee applied a structured five-factor decision framework:

### 7.1 Technical Completion Viability

At Month 10, despite the cost overruns, the core data architecture had been stabilised following the Month 8 redesign. The microservices skeleton was functional, and the team had gained sufficient competence in Kubernetes to maintain a stable deployment pipeline. The remaining work (integration testing, API completion, UAT) was better understood and lower risk than earlier phases. Technical feasibility assessment: viable with reduced scope.

### 7.2 Revised Financial Analysis

The sunk cost of \$1.1M was correctly excluded from the forward-looking analysis (sunk costs are irrelevant to future decisions). The decision was framed around the incremental investment required:

- Option A — Terminate: Sunk cost \$1.1M. Residual salvage value: approximately \$150,000 (reusable codebase components). Net loss: ~\$950,000. Legacy system must continue for 3+ additional years.
- Option B — Full Completion: Estimated additional cost \$1.4M (ETC at current CPI). Total project cost ~\$2.5M. Still cash-flow positive over a 4-year NPV horizon.

- Option C — Re-Scoped MVP Delivery: Reduce scope by 30% (defer AI forecasting, reduce API integrations to top 80 partners). Estimated additional cost \$780,000. Total ~\$1.88M. Break-even in 2.5 years.

### 7.3 Market and Strategic Window

FreightFlow's three largest clients had issued RFQs requiring real-time API integrations by Q3 2024. Failure to launch would risk losing contracts representing \$8.2M in annual revenue. The strategic window remained open, but was time-bounded. Each month of delay was estimated to cost \$120,000 in delayed contract conversions.

### 7.4 Stakeholder Confidence and Governance

The steering committee remained committed to digital transformation as a corporate strategy. Board approval for an additional \$900,000 was conditionally granted, subject to a revised project plan with weekly EVM reporting and defined corrective action triggers (CPI < 0.55 triggers mandatory review; CPI < 0.45 triggers termination assessment).

### 7.5 Decision Outcome

Option C — Re-Scoped MVP Delivery — was approved. The decision was documented as: "The project retains positive NPV with scope reduction, the strategic imperative remains valid, and technical risks are now better understood. Re-scoping is the rational choice, supported by EVM data."

## 8. Corrective Actions and Recovery Plan

### 8.1 Scope Rationalisation

A formal scope reduction exercise was conducted using MoSCoW prioritisation (Must Have, Should Have, Could Have, Won't Have). The following features were deferred to a Phase 2 release (September 2024):

- AI-powered demand forecasting module (deferred)
- API integrations for the bottom 120 of 200 vendors (deferred; top 80 retained)
- Advanced analytics dashboard (deferred)
- Mobile companion app (deferred)

Scope reduction budget saving: approximately \$320,000. Schedule saving: approximately 8 weeks.

### 8.2 Resource Acceleration

Four specialist contractors were onboarded: two senior cloud engineers with Kubernetes expertise, one integration specialist with documented experience in logistics API ecosystems, and one senior QA automation engineer. A dedicated QA environment (isolated AWS account) was provisioned, eliminating the shared infrastructure bottleneck.

### 8.3 Process and Governance Improvements

Formal EVM tracking was institutionalised with weekly reporting to the PMO. A Change Control Board was established, meeting weekly to evaluate all scope addition requests. A risk-adjusted EAC — incorporating a 12% contingency buffer — replaced the naïve BAC/CPI forecast as the primary management forecast. TCPI was tracked weekly against the revised BAC to ensure planned efficiency targets were realistic.

8.4 Post-Recovery Performance

Following corrective actions, Months 12–16 showed consistent improvement. By Month 16, the CPI had risen from 0.47 to 0.62 and the SPI stabilised at 0.98, indicating delivery closely aligned with the revised plan. The Phase 1 MVP launch was achieved in Month 20 (April 2024), four months ahead of the worst-case projection from Month 9.

9. Lessons Learned

9.1 The Cost of Delayed EVM Implementation

The CloudSync project demonstrates that EVM’s greatest value lies in its early warning capability. Had EVM been tracked from Month 1, the following consequences could have been avoided:

| Scenario                   | Without Early EVM              | With Early EVM (Month 3)                |
|----------------------------|--------------------------------|-----------------------------------------|
| Detection Point            | Month 10 (PMO intervention)    | Month 3 (EAC trend visible)             |
| Unnecessary Costs Absorbed | ~\$680,000 between Months 3–10 | Avoidable with early re-scoping         |
| Schedule Loss              | 7 months beyond original plan  | Estimated 3–4 months lost, recoverable  |
| Final Cost                 | ~\$1,880,000 (actual)          | Estimated \$1,450,000–1,550,000 (model) |

9.2 EVM as a Decision Support Tool

A critical lesson is that EVM does not make decisions; it informs them. At Month 9, a raw CPI of 0.47 might appear to suggest immediate termination. However, the full decision framework — incorporating NPV analysis, strategic value, technical feasibility, and stakeholder context — supported a different, ultimately more valuable outcome. EVM provided the factual foundation; judgment and business acumen provided the direction.

9.3 Change Control as a Pre-Condition for EVM Validity

EVM metrics are only meaningful if the performance measurement baseline is accurate and current. In the CloudSync project, uncontrolled scope additions corrupted the baseline, making CPI comparisons partially misleading until the baseline was formally re-established at Month 10. Change control is not merely a governance formality; it is a prerequisite for EVM integrity.

10. Conclusion

The CloudSync ERP Platform project provides a compelling illustration of both the cost of ignoring EVM signals and the value of introducing EVM — even belatedly — as a recovery instrument. When the project team reported ‘on track’ through Month 6 based on milestone completion narratives, the underlying EVM data told a starkly different story: a CPI declining to 0.65, an EAC forecast exceeding budget by 40%, and a TCPI climbing toward unrealistic territory.

EVM provided four distinct decision-support services throughout the project lifecycle. First, it served as an early warning system, with the Month 1–3 CPI trend alone sufficient to justify a project review. Second, it enabled objective forecasting, replacing optimistic narrative reports with mathematically grounded cost-at-completion estimates. Third, it structured the termination decision by quantifying the financial gap between continuation and termination options. Fourth, it validated the recovery, with post-intervention CPI and SPI trends confirming that corrective actions were producing measurable improvement.

The case reinforces a well-established finding in project management research: for projects of significant scale and complexity, EVM is not a luxury or an administrative overhead — it is a foundational governance mechanism.

Organisations that delay its implementation do not avoid the information; they simply avoid knowing what is already true about their project's trajectory until the cost of acting on that information has multiplied substantially.

For software projects in particular, where scope is fluid and technical uncertainty is high, the discipline of regularly computing EV and comparing it to both PV and AC creates an accountability structure that narrative reporting cannot replicate. The CloudSync project ultimately delivered its MVP eight months late and 57% over budget — a poor outcome by any measure. But it is a considerably better outcome than the \$2.5M, twelve-month-late scenario projected at Month 9, and the difference is attributable directly to EVM-driven decision making.

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## Appendix A: EVM Formulae Summary

| Metric | Formula                   | Interpretation                                                                   |
|--------|---------------------------|----------------------------------------------------------------------------------|
| CPI    | $EV / AC$                 | Cost efficiency. $< 1.0$ = over budget per unit of work                          |
| SPI    | $EV / PV$                 | Schedule efficiency. $< 1.0$ = behind schedule                                   |
| EAC    | $BAC / CPI$               | Projected total cost if current efficiency continues                             |
| VAC    | $BAC - EAC$               | Projected budget variance at completion                                          |
| ETC    | $EAC - AC$                | Remaining cost to complete the project                                           |
| TCPI   | $(BAC - EV) / (BAC - AC)$ | Efficiency required to finish on original budget. $> 1.10$ generally unrealistic |
| CV     | $EV - AC$                 | Cost variance. Negative = over budget                                            |
| SV     | $EV - PV$                 | Schedule variance. Negative = behind schedule                                    |

**Appendix B: Change Request Summary**

| CR# | Description                                                | Month   | Budget Impact | Schedule   | Status                       |
|-----|------------------------------------------------------------|---------|---------------|------------|------------------------------|
| 001 | Data localisation compliance (Malaysia/Indonesia)          | Month 4 | +\$92,000     | +7 weeks   | Approved – Mandatory         |
| 002 | Warehouse Management System integration scope expansion    | Month 6 | +\$45,000     | +3 weeks   | Informal – No change control |
| 003 | Security vulnerability remediation (authentication module) | Month 8 | +\$38,000     | +2 weeks   | Approved – Mandatory         |
| 004 | Performance tuning – database query optimisation           | Month 9 | +\$22,000     | +1.5 weeks | Informal – No change control |